

AKSHAY BAWALIWALE

Generative AI & Machine Learning Engineer | LLM Applications, Agentic Systems, RAG & Applied ML

+91 9826422264 akshay.aispecialist@gmail.com Website LinkedIn GitHub Medium

SUMMARY

Generative AI & Machine Learning Engineer with **2.5+ years** shipping **20+ production LLM applications: multi-agent systems** (LangGraph, CrewAI), **RAG pipelines**, and **Voice AI**. Built systems handling **5,000 requests/day** with **92% retrieval accuracy** and **50% cost reduction**. Equally strong in **applied machine learning** — building and benchmarking end-to-end systems from **recommendation engines** to **deep-learning computer-vision pipelines**.

TECHNICAL SKILLS

Languages: Python, R, SQL

GenAI and LLM: LangChain, LangGraph, CrewAI, LlamaIndex, Ollama, Hugging Face, RAG, Structured Output, Function Calling, Prompt Caching, Prompt Optimisation, Prompt Tuning, RLHF, PEFT, LLM Guardrails, Guardrails AI, Hallucination Reduction, Token Optimisation, LLM Observability, Agentic AI, MCP, Tool Calling, ReAct Agents

Vector Databases: FAISS, ChromaDB, Pinecone, Vector Store

ML Algorithms: Linear Regression, Logistic Regression, Decision Tree, XGBoost, KNN, Naive Bayes, Ensemble Methods, K-means Clustering, Anomaly Detection, Supervised ML, Unsupervised ML, Neural Networks

Cloud: AWS S3, Google Vertex AI, Azure

Observability and Testing: LangSmith, MLflow, A/B Testing, Model Evaluation and Benchmarking

Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn, NLTK

Dev Tools: Jupyter Notebook, Google Colab, VS Code, GitHub, Claude Code, Codex, AWS CodeCommit

Professional Skills: Prompt Engineering, Agentic Workflows, LLM Fine-Tuning, System Design, Scalability, Latency Optimisation, High Availability, Distributed Systems, Data Science, ML Engineering, Business Analytics, Predictive Modelling

EXPERIENCE

Senior Generative AI & LLM Specialist

Jan 2026 – Present · Remote

BuzzBoard

- Designed and deployed **IRIS**, a three-agent **Voice AI workflow** integrating **LiveKit, Twilio SIP, and Salesforce**, reducing request-queue latency from **two days to three minutes (99.9%)** and fulfillment time by **87.5%**, with real-time **HIPAA and PCI DSS compliance guardrails**.
- Led end-to-end **LLM evaluation and LLMOps** across response quality, latency, cost, and tool-use reliability using a **20-parameter evaluation framework** and **RAGAS**.
- Engineered production-grade **RAG content pipelines** that achieved **92% retrieval accuracy** and reduced manual review effort by **50%** through automated evaluation and quality-control workflows.
- Built and maintained **CI/CD pipelines on AWS** for scalable AI application deployments using **FastAPI, Git-based version control, and MLflow**, enabling continuous performance, experiment, and cost monitoring.
- Mentored **three engineers** across agentic system design, production deployment, evaluation frameworks, and engineering best practices.

Generative AI & LLM Specialist

Apr 2024 – Jan 2026 · Remote

BuzzBoard

- Architected and delivered **20+ production-grade Agentic AI systems** for BuzzBoard's US SMB client base, including **multi-agent BrandKit generation, Voice AI customer service, conversational agents, and social-content pipelines**, owning the complete lifecycle from architecture and development to deployment and production handover.
- Built a **multi-agent B2B sales-intelligence platform** using **ReAct-based Search, Knowledge, and Web agents**, reducing prospect research and lead-conversion cycles from **months to days** and cutting manual research effort by **75%**.

Full stack Data Science Trainee

Apr 2023 – Mar 2024 · Remote

AlmaBetter

- Completed an **Advanced Full-Stack Data Science and Machine Learning** program, with **800+ hours** of hands-on coding across **Python, SQL, EDA**, data preprocessing, **statistical analysis**, **feature engineering** and **predictive modelling**.
- Delivered **50+ end-to-end data science and machine learning projects** and assignments, covering problem formulation, data cleaning, visualization, **model development**, **hyperparameter tuning**, performance evaluation and business insight generation.

Research Staff

Jul 2022 – Oct 2022 · Remote

Centre of Excellence in AI, IIT Kharagpur

- Developed **image-processing and preprocessing algorithms** and contributed to **satellite imagery annotation** for a **U-Net-based semantic segmentation pipeline**, improving model accuracy by **86%**.

FEATURED PROJECTS

Netflix AI Recommender System [GitHub](#)

Python, scikit-learn, Flask, NMF, LDA

- Built a **Machine Learning-based Netflix recommendation system**, benchmarking **11 models** across the complete catalog of **8,807 titles**. Deployed a hybrid recommendation model that achieved **99.96% Precision@10**, improved content diversity by **43%**, and delivered recommendations in **6.91 ms**.
- Identified and eliminated a high-accuracy model that created a repetitive **genre echo chamber**, prioritizing balanced and relevant recommendations over a single performance metric. Added **cold-start recommendations** for new users and **explainable results** for greater transparency.

AI-Powered Image Super-Resolution API [GitHub](#)

PyTorch, Real-ESRGAN, Flask

- Engineered and deployed a production-ready image super-resolution API using **Real-ESRGAN** and **RRDBNet**, delivering **4x image upscaling** while preserving fine visual details. Optimized inference for resource-constrained environments using **tile-based processing**, **FP16 acceleration**, **thread-safe model loading**, and **automatic GPU/CPU fallback**.

EDUCATION

M.Tech, Earth System Science and Technology – IIT Kharagpur (2019–2021)

B.E., Mechanical Engineering – SSGI, Bhilai (2012–2016)

CERTIFICATIONS & WRITING

Reinforcement Fine-Tuning LLMs with GRPO – DeepLearning.AI

Machine Learning Specialization – Stanford University & DeepLearning.AI

Technical articles on **generative AI**, **agentic engineering**, and **LLM reinforcement learning** – Medium: [@akshay.bawali09](#)